

StyleVector

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A reproduction of personalized text generation via activation steering

PROBLEM STATEMENT LLMs write in one average style, yet real use needs a person's own voice. RAG slows down as history grows and PEFT needs per-user training. **StyleVector** captures a user's style as one vector in activation space, with no training.

NOVELTY

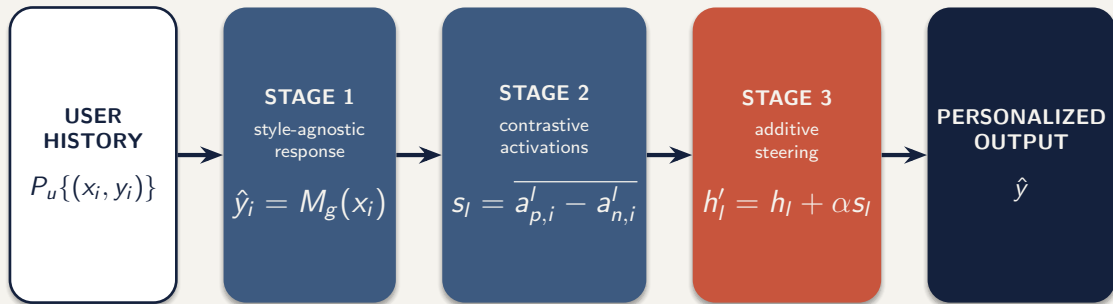
- 01 From-scratch reimplementation
- 02 Live demo: baseline vs. steered
- 03 Planned: top- k PCA + norm-preserving steering

CONTRIBUTIONS

Solo project

Slok Tulsyan (12342080)

- Reimplemented Eq. 1–4 in PyTorch
- Evaluated 1,000 users on LaMP-4/5/7
- Built the live demo and the report



Live Demo: <http://10.10.0.247:5000/>

Results and Findings

StyleVector

Task	Metric	Paper			Ours (n = 1,000)		
		Base	StyleVector	Δ	Base	StyleVector	Δ
LaMP-4	ROUGE-L	0.0398	0.0411	+3.3%	0.0988	0.1020	+3.3%
LaMP-4	METEOR	0.0790	0.0809	+2.4%	0.1172	0.1207	+3.0%
LaMP-5	ROUGE-L	0.1086	0.1366	+25.8%	0.1708	0.2102	+23.1%
LaMP-5	METEOR	0.2337	0.2575	+10.2%	0.2829	0.3125	+10.5%
LaMP-7	ROUGE-L	0.2506	0.2827	+12.8%	0.2525	0.2942	+16.5%
LaMP-7	METEOR	0.2588	0.3042	+17.5%	0.2770	0.3239	+16.9%

StyleVector beats the baseline on all 6 task–metric pairs, as in the paper.